

returns, the ARCH model will apply greater weight to the more recent observations. This allows the model to update quickly in a changing volatility environment.

A simple ARCH(1) model consists of the previous day's price return. This is formulated as:

$$\hat{\sigma}_t^2 = \omega + \alpha_1 y_{t-1}^2 \quad (6.19)$$

where, $\omega > 0, 0 \leq \alpha_1 < 1$.

GARCH Volatility Model

The GARCH volatility model was introduced by Bollerslev (1986) and is an extension of the ARCH model (Engle, 1982). A GARCH(p,q) model consists of "p" previous returns and "q" previous volatility forecasts as follows:

$$\hat{\sigma}_t^2 = \omega + \sum_{i=1}^p \alpha_i r_{t-i}^2 + \sum_{j=1}^q \beta_j \hat{\sigma}_{t-j}^2 \quad (6.20)$$

where, $\omega > 0, \alpha_1, \dots, \alpha_p, b_1, \dots, b_q \geq 0, \sum \alpha_i + \sum \beta_j < 1$.

The GARCH model applies more weight to the more recent observations thus allowing the model to quickly adapt to changing volatility regimes. The parameters of the model are determined via maximum likelihood estimation.

A simple GARCH(1,1) model consists of only the previous day's price return and previous day's volatility forecast and is formulated as:

$$\hat{\sigma}_t^2 = \omega + \alpha_1 y_{t-1}^2 + \beta_1 \hat{\sigma}_{t-1}^2 \quad (6.21)$$

where, $\omega > 0, \alpha_1, b_1 \geq 0, \alpha_1 + b_1 < 1$.

HMA-VIX ADJUSTMENT MODEL

The HMA-VIX volatility forecasting model is an approach that combines the stock's current volatility with an implied volatility estimate. We formulate this model as:

$$\hat{\sigma}_t = \bar{\sigma}_{t-1} \cdot \frac{VIX_{t-1}}{\bar{\sigma}_{SPX,t-1}} \cdot AdjFactor \quad (6.22)$$

where $\bar{\sigma}_{t-1}$ is the HMA stock trailing volatility, $\bar{\sigma}_{SPX,t-1}$ is the SP500 index trailing volatility, VIX_{t-1} is the VIX implied volatility index, and *AdjFactor* is the adjustment factor needed to correct for the risk premium embedded in the VIX contract.